Minyong Jeong

Github LinkedIn
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I focus on Embedded Software and Edge AI, with interests in sensor fusion, on-device inference, and NPU deployment. My recent projects include an STM32F4 based autonomous maze navigation system and an EO-IR sensor fusion object detection architecture ported to the DX-M1 NPU for edge-device monitoring.

I am especially interested in applying AI technologies to solve real-world problems. Through projects that connect embedded systems, edge devices, and AI models, I aim to build practical systems that can operate reliably in real environments.

SKILLS

Languages Python · C | Korean (Native) · English (TOEIC Speaking 130, IM3)
Embedded SW FreeRTOS · STM32F4 · Raspberry Pi · NPU Deployment
AI PyTorch · YOLO · Object Detection · EO-IR Sensor Fusion

PROJECTS

Embedded-Challenge
April 2026 - June 2026

- Designed a FreeRTOS-based task architecture that separated sensor processing and driving control, using Queue-based inter-task communication to improve modularity and real-time data flow.
- Developed a state-based navigation algorithm using wall-detection history to resolve open-space misclassification issues and improve driving stability in autonomous maze navigation.

DroneGuard-Edge
March 2026 - June 2026

- Deployed an EO-IR object detection model on the DeepX DX-M1 NPU and built a real-time inference pipeline for edge-device environments.
- Implemented a WebSocket-based video streaming and detection-result transmission system on Raspberry Pi 5, enabling real-time monitoring of inference outputs.
- Received the Excellence Award at the 2026 CNU SW/AI Project Fair, ranking 3rd out of 47 teams.

PUBLICATIONS

Environment-Conditioned Hammering-Precursor Monitoring for Reliable Embedded Systems
Korea Computer Congress (KCC) 2026, Poster Session — accepted
***Minyong Jeong**, Joonmo Jeong, Woojin An, Seohyeon Cho, Seungwoo Jeong, Venkatesan Muthukumar(University of Nevada, Las Vegas)*

- Proposed an environment-conditioned hammering-precursor monitoring model for embedded systems by fusing HPC time-series data with environmental signals such as core voltage and temperature.
- Designed a feature-level fusion architecture that separately processes performance-counter data and environmental conditions to improve robustness under unseen operating conditions.
- Evaluated the proposed method under out-of-distribution conditions, including high-temperature and low-voltage environments, achieving 99.57% accuracy and 99.60% F1-score with the InceptionTime-based model.

Reinforcement Learning-based Enhancement of Navigation and Dynamic/Static Obstacle Avoidance for Autonomous Surface Vehicles
Korea Software Congress (KSC) 2025, Poster Session
***Minyong Jeong**, Minjae Kim, Youngjun Kim, Jihun Park*

- Proposed an RL-based navigation and obstacle avoidance method for autonomous surface vehicles in complex coastal and port environments with dynamic and static obstacles.
- Built a Unity-based ASV simulation environment and outperformed A* and RRT*, reducing arrival time by 34–52%, and travel distance by 3–14%.

AWARDS

Excellence Award
2026 CNU SW/AI Project Fair

- Awarded 3rd Place among 47 teams at the 2026 CNU SW/AI Project Fair for the DroneGuard-Edge project.

Multiple Awards
CNU DevDay Programming Contest

- Ranked 3rd out of approximately 80 participants in the CNU DevDay Algorithm Contest.
- Also received three Silver Awards and one Bronze Award in CNU DevDay contests from 2024 to 2026.

EDUCATION

B.S. in Computer Science and Engineering, Minor in Cybersecurity.
Chungnam National University - Daejeon, Republic of Korea

03/2021 - 02/2027 (Expected; coursework requirements completed; available full-time from Sep 2026) | GPA: 4.24 / 4.50, Major GPA: 4.39 / 4.50